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Critical feedback impacts creative ideation and brain oscillations

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Abstract

Creative thinking is a vital skill for engineers. Prior work suggests that social dynamics—such as critical feedback from a high-authority figure—can influence the ideation process. Yet little is known about the neurocognitive mechanisms through which feedback shapes creative thinking. In this study, engineering students completed a creative ideation task while EEG was recorded. Midway through the experiment, a professor gave the participant either supportive or unsupportive critical feedback on their performance. Supportive feedback was expected to positively influence creativity compared to unsupportive feedback: participants in the supportive feedback condition were predicted to show greater idea originality and fluency after receiving feedback, as well as a greater EEG power increase in the alpha frequency band (8-12 Hz) that is robustly associated with creativity. We found that after receiving feedback—whether supportive or unsupportive—participants produced fewer but more highly original responses and showed increased alpha power. These results indicate that feedback can cause engineers to generate fewer but more original ideas by driving alpha-band activity in the brain. In further analyses, we found decreased beta-band activity before feedback only in the unsupportive condition, possibly reflecting increased cognitive stress and internally directed attention required to adjust performance in the post-feedback phase.

Keywords: EEG, alpha, oscillations, creativity, feedback

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The ability to think creatively is vital for success in a variety of careers. Engineers, for example, must use creative divergent thinking (i.e., the process of generating novel and useful ideas) to produce innovative engineering design ideas. Divergent thinking in workplaces and classrooms often takes place in group settings, making the creative process susceptible to the influence of social factors such as critical feedback from high authority individuals or role-models. Prior work has used electroencephalography (EEG) neural oscillations to understand the cognitive processes underlying creative ideation, generally finding that creative ideation is associated with increases in oscillation power in the alpha (8-12 Hz) frequency band (i.e., Benedek et al., 2011; Fink & Neubauer, 2006; Jauk et al., 2012; Jończyk et al., 2022; Rominger et al., 2019; Schwab et al., 2014). Building on this work, the present study investigates the impact of critical feedback on creative ideation and the underlying neural processes, aiming to better understand the neurocognitive mechanisms through which feedback shapes creative thinking.

1.1 Influence of Feedback on Creativity

Creative thinking is a vital skill for engineers but one that is difficult to teach in classrooms. Engineering educators, for example, face the challenge of teaching engineering students how to think creatively, specifically how to look past their initial ideas and continue thinking of novel solutions—a skill necessary for success in the engineering workforce and continued innovation in the field. Most engineering analysis courses emphasize finding a single correct solution, whereas engineering design courses prioritize creativity, including the ability to understand the design process and to engage in divergent thinking or ideation—that is, generating multiple novel solutions to a single problem. However, in many engineering programs, the limited number of design courses compared to analysis courses means that students often lack sufficient opportunities to practice and develop their ideation skills, which are essential for generating innovative engineering solutions. A lack of creative ability in the engineering field has been identified as a problem both empirically by researchers (Klukken et al., 1997; Leon-Rovira et al., 2008; Masters et al., 2008) and in popular opinion (i.e. Forbes, Banerjee, 2014; TIMES, White, 2013). Furthermore, creativity is not always encouraged in engineering environments. Undergraduate students, for example, have been found to perceive their academic environment as increasingly less supportive of creative innovation throughout their undergraduate career, and the originality of their creative ideas decreased in this same time period (Masters et al., 2008).

In engineering education (and in other educational and workplace settings) ideation and design processes often occur in group settings or classrooms which makes them susceptible to the positive or negative influence of other individuals and social dynamics (e.g., the invocation of negative stereotypes; Jończyk et al., 2022). Specifically, creativity may be impacted by feedback from high authority figures such as professors. Research has found that role-models (individuals who have attained success in their field whom younger members of the field look up to) can both positively and negatively influence the interest and success of students. For example, students in STEM rely on role-models for positive feedback and approval (Seron et al., 2016), and a lack of female role models may discourage women from STEM participation (Cheryan et al., 2017). However, research finds that role-models must also be relatable (Cheryan et al., 2017). For example, when female role-models invoked negative stereotypes about computer scientists (thus unrelatable), female participants *reduced* their interest in majoring in computer science (Cheryan et al., 2013).

In addition to the relatability of role-models, the type and quality of feedback they give may also influence student performance on creativity and tasks requiring creative ideation. According to the feedback intervention theory, feedback on creative performance makes an individual aware of a gap between their performance and a "standard" (Kluger & DeNisi, 1996). Individuals can react to this feedback in one of two ways. Feedback can either encourage individuals to generate and implement better task strategies (thus leading to improved creative performance) or an individual may feel threatened by the feedback and be less willing to engage in creative ideation (Kim & Kim, 2020; Kluger & DeNisi, 1996). Overly critical feedback can negatively affect students' performance. However, "wise" supportive feedback—in which the role-model communicates their high standards and confidence in the students' ability to achieve them—motivates students to address criticism (G. L. Cohen et al., 1999; Yeager et al., 2014). Supportive feedback can thus allow mentors to give critical advice while keeping a student's trust, potentially leading to improved performance. The present study will investigate the role of supportive critical feedback—relative to unsupportive feedback—on engineering students' creative ideation.

1.2 EEG Oscillations and Creativity

EEG has been used to investigate the cognitive processes underlying creative ideation—it is highly sensitive to neural activity at the temporal level, making it an ideal technique to study processes such as creative ideation. EEG power analysis in the time-frequency domain can be used to compare task-related EEG activity to that of a preceding reference resting interval, observing either power increases in activity from the reference (event-related synchronization; ERS) or decreases in activity from the reference (event-related desynchronization, ERD). The alpha frequency band (8-12Hz) is generally associated with inhibitory functions (Klimesch,

2012) and has been found to be sensitive to a variety of cognitive processes including memory (e.g., Klimesch, 1999), language processing (e.g., Bakker et al., 2015; Bastiaansen et al., 2012; Liu & Van Hell, 2024), metaphor generation (Yu et al., 2025) and, importantly, creative ideation (e.g., Benedek et al., 2011; Fink et al., 2009; Fink & Neubauer, 2006; Händel et al., 2024; Jauk et al., 2012; Jończyk et al., 2022, 2024; Rominger et al., 2019; Rominger, Gubler, et al., 2022; Rominger, Benedek, et al., 2022; Schwab et al., 2014; Stevens & Zabelina, 2019; Yin et al., 2024). Specifically, synchronization (ERS) within the alpha band is typically observed in tasks requiring high levels of internal attention and suppression of external input, including in tasks requiring creativity (Benedek et al., 2011, 2014; Benedek, 2018).

To investigate power changes during creative ideation, previous studies have employed a technique in which participants give verbal responses to creativity prompts, with button presses demarcating responses, and then analyze brain oscillations just before ideas are given (i.e., Benedek et al., 2011; Fink & Neubauer, 2006; Jauk et al., 2012; Jończyk et al., 2022; Rominger et al., 2019; Schwab et al., 2014). These findings consistently indicate that creative ideation is associated with increased alpha power over bilateral frontal sites and right-hemisphere posterior sites (Benedek et al., 2011; Fink & Neubauer, 2006; Jauk et al., 2012; Jończyk et al., 2022; Rominger et al., 2019; Rominger, Gubler, et al., 2022; Schwab et al., 2014). Alpha increases are observed not just from reference to active creative ideation periods but also during the generation of more creative ideas compared to less creative ones (Fink & Neubauer, 2006; Rominger, Gubler, et al., 2022), during periods of mind-wandering (Compton et al., 2019), during idea evaluation compared to idea generation (Hao et al., 2016), and as a function of individual differences in creativity, openness to experience, or metacognitive monitoring (Fink et al., 2009; Fink & Benedek, 2014; Martindale & Hasenfus, 1978; Rominger, Benedek, et al., 2022).

Within the alpha frequency range, the upper alpha (10-12Hz) band is typically thought to be sensitive to specific task requirements, while the lower alpha (8-10Hz) band is thought to be sensitive to general task demands (Benedek et al., 2011; Benedek, 2018; Klimesch, 1999). ERS during creative ideation has been observed in both the upper and lower alpha bands, and results are typically similar or highly correlated between the two sub-bands (i.e., Benedek et al., 2011; Fink et al., 2009; Fink & Neubauer, 2008; Jauk et al., 2012; Jończyk et al., 2022, 2024; Rominger et al., 2019; Schwab et al., 2014). However, some studies have found particularly pronounced effects in the upper alpha band, for instance observing U-shaped changes in upper alpha synchronization over time (i.e., greater synchronization at the beginning and end of a creative ideation period) as well as upper alpha functional coupling between frontal and posterior/occipital electrodes (Rominger et al., 2019; Rominger, Gubler, et al., 2022). Increases in alpha power observed during creative ideation tasks (particularly over right-hemisphere posterior sites) are typically thought to reflect high levels of internally directed attention that are required during creative ideation (Benedek et al., 2011; Rominger, Gubler, et al., 2022).

As creative ideation often occurs in group settings, it may thus be susceptible to the influence of social dynamics. A small number of studies have examined the influence of social dynamics on alpha synchronization during creative ideation (e.g. Fink et al., 2011; Jończyk et al., 2022). For example, Fink et al. (2011) found that alpha synchronization was greater during creative ideation when participants were first exposed to other peoples' creative ideas compared to a no-exposure control. Furthermore, Jończyk et al. (2022) found that stereotype threat invocation affected alpha power in the creative ideation of women engineers. Alpha synchronization increased after the female participants were reminded by a confederate, midway through the experiment, of the stereotype that women are not good at creative thinking. Although the creativity of behavioral responses did not change after the stereotype threat, at the neural level, alpha power increased, which was interpreted to mean that the participants used more internally directed attention and exerted greater effort on the task after the stereotype threat. The effect of another potentially important factor affecting creative thinking—critical feedback—on alpha power changes during creative ideation as well as behavioral creative outcomes has yet to be investigated.

1.3 Present Study

The present study investigates the effect of critical feedback from a high-authority figure on creative ideation performance in engineering students. While EEG was recorded, participants completed two divergent creative thinking tasks——the Alternate Uses Task (AUT; Guilford, 1967) and the Utopian Situations Task (UST; Wilson et al., 1954). In the AUT, participants were asked to generate alternate, unusual, or creative ways to use everyday objects (e.g., a brick). In the UST, participants were given an imaginary situation and asked to generate ideas of what might happen in that situation (e.g., What would happen if nobody could speak anymore?). For both tasks, participants were asked to generate as many and as creative responses as possible within the 2-minute time frame per item. These two tasks were used to ensure that creative ideation was being examined broadly and was not restricted to the nuances of any one creativity task. EEG alpha power changes were measured in the time interval just before each idea was given aloud and compared to a reference period before each trial.

Critically, mid-way through the creative ideation experiment, a professor entered the recording booth and gave scripted supportive or unsupportive critical feedback to the participant (see Appendix A for the full scripts). While both individuals gave critical feedback, the supportive professor additionally expressed their high standards for the participant and their belief in the participant's ability to meet those standards, known as "wise feedback" (G. L. Cohen et al., 1999; Yeager et al., 2014). Wise feedback effectuates a higher motivation in students to address the constructive criticism and improve the assignment (e.g., G. L. Cohen et al., 1999; Yeager et al., 2014). Not only has wise feedback been found to be more effective than regular (unbuffered) critical feedback or overpraised feedback (e.g., Biernat & Manis, 1994; Brummelman et al., 2014) in motivating students to improve their performance, it also expresses trust in a student's capacity that allows students to interpret criticism as an opportunity for further improvement. We predicted that this supportive feedback, more so than unsupportive feedback, would positively impact creative performance, as reflected in idea fluency (the number of ideas produced) and idea originality (creativity of responses). In turn, we also predicted that alpha increases from the pre-feedback to post-feedback trials would be greater for the supportive feedback group compared to the unsupportive feedback group.

2 Methods

2.1 Participants

Sixty-seven undergraduate and graduate engineering students from a large American university participated in this study. All participants gave informed consent, and the experiment was approved by the university's IRB. Of these, eleven participants were excluded due to technical issues with EEG recording and two due to not completing all tasks. A total of 54 participants were included in the final sample (M_{age} = 20.96, SD = 2.33; 19 women, 35 men). All participants self-reported being right-handed with normal or corrected to normal vision and no history of language or neurological disorders. Additionally, all participants were fluent in English, with 31 participants reporting a first language other than English including Spanish, Hindi, and Chinese. Of the final sample of 54 participants, 28 participants received supportive feedback while 26 received unsupportive feedback.

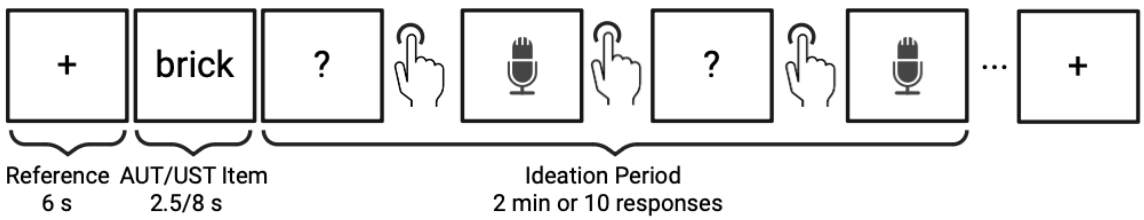
2.2 Experimental Tasks

The Alternate Uses Task (AUT) and Utopian Situations Task (UST) were completed while EEG data was recorded, with order counter-balanced across the feedback conditions. In the AUT, participants were asked to generate creative, original, or novel uses for eight everyday objects. Half of the objects were general (brick, pencil, hanger, and key) and half were engineering-related (foil, magnet, helmet, and pipe). For the UST, participants were asked to generate ideas for what would happen in eight hypothetical situations. Half of the situations were general (e.g., What would happen if nobody could speak anymore?) and half were engineering-related (e.g., What would happen or might be changed, if there was no gravity in the world?). Both tasks were presented using E-prime 2.0 (Psychology Software Tools, Inc.).

Procedures for these two tasks were based on previous work using the AUT and UST with EEG (Fink & Neubauer, 2006; Jończyk et al., 2022). As depicted in Figure 1, each test item began with a six-second pre-stimulus fixation which served as the baseline reference periods for later analysis. Then, the prompt was shown on the screen for 2.5 seconds (AUT prompts) or 8 seconds (UST prompts) before being replaced by a question mark. Participants silently generated ideas in their heads in a self-paced design. When an idea came to mind, the participant pressed the middle button on a button box, spoke the idea aloud, then pressed the button again. The question mark returned to the screen and participants again silently generated their next idea, repeating this process for two minutes or until a maximum of ten responses were given for a particular item. The next trial then began after another 6 second fixation. Responses were recorded using a Marantz PMD661 recorder.

Figure 1

Schematic of EEG experimental procedure.



2.3 Procedure

Participants first completed language and education history questionnaires and the Edinburgh Handedness Test (Oldfield, 1971) then were seated in a dimly-lit sound-attenuated booth roughly 100 cm from a computer screen. Participants were randomly assigned to either the supportive feedback or unsupportive feedback condition. During the EEG cap preparation, two researchers performed a scripted conversation, describing interactions with an academic mentor with either a supportive or unsupportive feedback style (see full script in Appendix B). Additionally, for the supportive feedback condition, photos of the professor in group photos with their students were posted around the lab. No pictures were present for the unsupportive feedback condition.

First, pre-experimental EEG resting state data was recorded: two minutes with the participants' eyes open and two minutes with their eyes closed. Next, instructions were given for the AUT and UST tasks, and one practice item was completed for each task. The experimental task was completed as follows. First, participants completed one block of each task (four prompts in each block, two of which were general and two of which were engineering-related;

presentation order was randomized within each block) in a counterbalanced order.[1] Then, at the halfway point of the experiment, a professor entered the room and gave pre-scripted feedback to the participant on their performance. One of two feedback types was given to the participant: supportive feedback in which the (female) professor told the participant that they had provided creative ideas so far and encouraged them to continue to do so, or unsupportive feedback in which the (male) professor told the participant that their ideas had been "basic", and they should try harder. These conditions represent the stereotypically extreme ends of the spectrum. See Appendix A for full scripts. After the professor left, the participant completed an additional block of each task, in the same order as the first two blocks. Finally, participants again completed post-experiment eyes-open and eyes-closed resting-state recordings for two minutes each.

After the EEG experiment, participants completed additional behavioral measures including a post-task survey, the AX-CPT measure of cognitive control (J. D. Cohen et al., 1999; Rosvold et al., 1956), the Operation Span task (Turner & Engle, 1989), and the Boston Naming Task (Kaplan et al., 1983). The post-task survey included a self-efficacy scale (Bandura et al., 1999), the Big Five Inventory (Goldberg, 1992), and a manipulation check. The manipulation check included six statements regarding participants' perceptions of the professor and the mid-experiment feedback (i.e., the professor who came in mid-experiment was agreeable, intelligent, relatable, supportive, motivating, and helpful; see Appendix C for full survey questions).

[1] As we had no *a priori* predictions regarding prompt type, it is not included in our main analysis. A supplementary analysis of both behavioral and EEG data including prompt type as an additional factor is reported on our OSF repository.

Participants rated each statement on a scale from 1 (strongly disagree) to 7 (strongly agree), and these responses were summed for a single score for each participant. After completing all tasks, participants were debriefed about the aim of the experiment and the nature of the feedback intervention.

2.4 Behavioral Scoring and Analysis

Responses to the AUT and UST tasks were transcribed by native English speakers then scored on two measures: originality and fluency. Response originality for each item was evaluated using Open Creativity Scoring with Artificial Intelligence (Ocsai Version 1.6, Organisciak et al., 2024), a large language model trained on scoring a variety of creativity tasks (including Alternate Uses and Utopian Situations tasks) whose ratings highly correlate with human-scored originality ratings (Organisciak et al., 2024). Each response was given a score ranging from 1 (unoriginal) to 5 (very original). A single originality score was calculated for each participant, as the average of the originality across prompts. Fluency was calculated as the number of responses for each prompt, averaged for each participant.

Behavioral analyses were conducted in R (Version 4.4.1, 2024). Scores for AUT and UST were combined for a single measure of creativity. Originality and fluency scores were analyzed separately, in mixed ANOVAs with FEEDBACK TYPE (supportive feedback, unsupportive feedback) as a between-subjects factor and PHASE (pre-feedback, post-feedback) as a within-subject factor.

2.5 EEG Acquisition

Continuous EEG data was recorded from 30 active Ag/AgCl electrodes (BrainVision Products GmbH, Germany) on an extended 10-20 convention (Jasper, 1958; Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2, PO10), referenced online to electrode FCz and continuously sampled at a rate of 500 Hz. Electrode impedances were kept below 5 kΩ. Signals were amplified using Neuroscan SynAmps2 (El Paso, TX) with a bandpass filter between 0.05 and 200 Hz.

Preprocessing steps were performed using Matlab R2023 (The MathWorks, Inc.). Data was bandpass filtered between 1 Hz and 55 Hz and manually cleaned for large unsystematic muscle artifacts by removing segments of poor-quality data. Using visual inspection, some channels were found to contain continuous low-quality data. These bad channels were removed (M = 1.19, *min* = 0, *max* = 3). Data was re-referenced to the algebraic mean of the left and right mastoid. Vertical and horizontal eye channels were excluded from further processing or analysis, and Independent Component Analysis (ICA) was performed using *runica* in EEGLAB. Independent components containing over 70% eye movement or 90% channel noise were removed (M = 2, *min* = 1, *max* = 4). Previously removed bad channels were then interpolated using the spherical method in EEGLAB.

Following Jończyk et al. (2022), task related power (TRP) was computed from the middle 4000 ms within the 6000 ms reference interval and 4000 ms around each button press (-3000 to 1000) of the activation intervals. Time/frequency decomposition was applied to both the activation and reference intervals using sinusoidal wavelet transforms (newtimef function in Matlab; wavelet scale expansion factor of 0.8), with 3 cycles at the lowest frequency (2 Hz), increasing linearly up to 22.5 cycles at the highest frequency (30 Hz). To establish the changes in the activation periods relative to the power during the baseline period (reference interval), following M. X. Cohen (2014), percentage change values were calculated at each time-frequency point at an electrode relative to baseline power:

$$\text{prctchange}_{\text{tf}} = 100 \cdot \frac{\text{activity}_{\text{tf}} - \text{baseline}_{\text{f}}}{\text{baseline}_{\text{f}}}$$

Thus, negative TRP percentage values indicate alpha power decrease from the baseline to the activation period (i.e., event-related alpha desynchronization; ERD); positive TRP percentage values indicate alpha power increase from the baseline to the activation period (event-related alpha synchronization; ERS).

2.6 EEG Analysis

EEG analyses focused on task-related power changes from the reference interval to the critical activation ideation period from -1500 ms to -500 ms before button presses. We conducted analyses on the right-hemisphere posterior region of interest (ROI: CP2, CP6, P4, P8, PO10, and O2) based on previous findings relating creative ideation to robust ERS in the right-hemisphere posterior region (i.e., Benedek et al., 2011; Fink & Benedek, 2014; Jończyk et al., 2022, 2024; Schwab et al., 2014).[2] TRP values were analyzed using a Mixed ANOVA comparing FEEDBACK TYPE (supportive feedback, unsupportive feedback) as a between-subjects factor and PHASE (pre-feedback, post-feedback) as a within-subjects factors. When applicable, Greenhouse-Geisser corrections were applied, and *p*-values obtained from post-hoc comparisons were adjusted using the Tukey correction. Separate analyses were conducted for the two frequency bands of interest: lower (8-10 Hz) and upper (10-12 Hz) alpha. Additionally,

[2] Supplementary analyses that include Hemisphere (right, left) and Area (anterio-frontal, fronto-central, centro-temporal, centro-parietal, parietal, parieto-occipital) as additional factors are reported on our OSF repository. The general patterns of effects for the variables of interest (Phase and Intervention) are comparable across the ROI and full-scalp approaches.

exploratory analyses were conducted for theta (4-7Hz), lower beta (12.5-16.5Hz), mid beta (16.5-20Hz), and upper beta (20-28Hz) frequency bands across all electrodes.

3 Results

3.1 Manipulation Check

Participants in the supportive feedback condition rated the professor significantly higher on the summed agreeable, intelligent, relatable, supportive, motivating, and helpful score (M = 31.64) than participants in the unsupportive feedback condition (M = 24.35, $t(214) = 10.82$, $p < .001$, 95% CI [-8.63, -5.97]).

3.2 Behavioral Results

For the idea originality ANOVA, there was a significant main effect of PHASE ($F(1, 52) = 17.34$, $p < .001$, $\hat{\eta}_p^2 = 0.25$) such that originality was higher post-feedback (M = 2.34, 95% CI [2.27, 2.41]) than pre-feedback (M = 2.22, 95% CI [2.15, 2.29]). For the idea fluency ANOVA, there was also a significant main effect of PHASE ($F(1, 52) = 10.79$, $p = .002$, $\hat{\eta}_p^2 = 0.17$), such that fluency was lower post-feedback (M = 5.83, 95% CI [5.27, 6.38]) than pre-feedback (M = 6.31, 95% CI [5.81, 6.82]).

Overall, behavioral creative thinking outcome measures indicate that post-feedback, regardless of whether the feedback was supportive or unsupportive, participants produced fewer but significantly more original ideas. These behavioral results are consistent with the serial order effect in creativity (e.g. Beaty & Silvia, 2012). To further test the interpretation that increased originality and decreased fluency were a result of the feedback intervention rather than just a function of time-on task, we used linear models to investigate changes in originality and fluency across the four experimental blocks. Recall that blocks one and two occurred before the feedback; blocks three and four occurred after the feedback. We first calculated the change in originality and fluency from block two minus block one (pre-feedback blocks), block three minus block two (feedback adjacent blocks) and block four minus block three (post-feedback blocks). We used orthogonal contrast codes for the three levels to test for linear (-1, 0, 1) and quadratic effects (-1, 2, -1). For idea originality, there was neither a significant linear (F(1, 159) = 0, p = .98, $\hat{\eta}_g^2 = 0$) nor quadratic effect (F(1, 159) = 0.03, p = .87, $\hat{\eta}_g^2 = 0$). For idea fluency we observed a significant quadratic effect (F(1, 159) = 6.46, p = .01, $\hat{\eta}_g^2 = 0.04$) but not a signification linear effect (F(1, 159) = 0.54, p = .46, $\hat{\eta}_g^2 = 0$). There was a greater decrease in idea fluency from the block preceding to the block following the feedback (M = -0.52, 95% CI [-0.88, -0.16]) compared to the two pre-intervention or two post-intervention blocks (M = 0.04, 95% CI [-0.21, 0.29]). Therefore, the observed decrease in idea fluency from pre-feedback to post-feedback above can be attributed to the feedback, rather than just time-on-task.

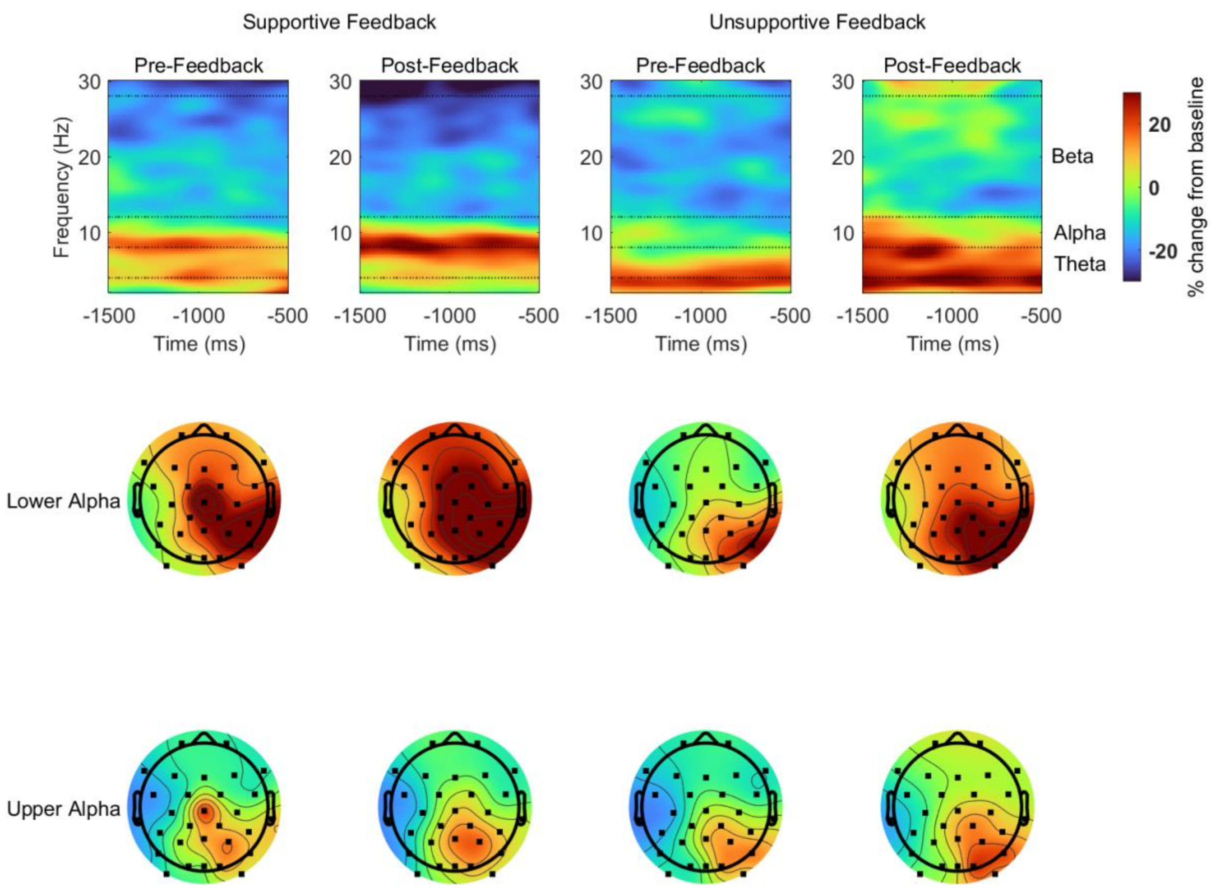
3.3 Electrophysiological Results: Lower and Upper Alpha

Mixed ANOVAs were conducted for the lower and upper alpha frequency bands with factors FEEDBACK TYPE and PHASE. In the upper frequency band (10-12Hz), we observed alpha synchronization (as seen in Figure 2) however, no main effects or interaction between FEEDBACK TYPE or PHASE reached significance. In the lower alpha frequency band (8-10Hz), a significant main effect of PHASE (F(1, 52) = 16.54, p < .001, $\hat{\eta}_p^2 = 0.24$) showed that TRP was greater post-feedback (M = 39.54, 95% CI [21.6, 57.48]) than pre-feedback (M = 23.64, 95% CI [9.86, 37.43]). Task related power and scalp maps for both the upper and lower alpha bands can be seen in Figure 2.[3]

[3] For all analyses, full model summaries can be found on our OSF repository.

Figure 2

Task-related power changes across all electrodes pre- and post-feedback for the supportive and unsupportive feedback groups. Scalp maps reflect topographical distribution of the effects in the -1500 to -500 ms time window for lower (8-10Hz) and upper (10-12Hz) frequency bands.



As with behavioral data, serial order effects have been found in alpha oscillations during creative processing——EEG alpha power increases over time during creative ideation (Agnoli et al., 2020). To further test the interpretation that alpha TRP increases observed from pre- to post-feedback were related to feedback, rather than a function of time-on-task, we again used a linear model to investigate changes in TRP across the four experimental blocks, using orthogonal contrast codes for the three levels to test for linear (-1, 0, 1) and quadratic effects (-1, 2, -1).

In the lower alpha band, we observed a significant quadratic effect ($F(1, 159) = 6.66$, $p = .01$, $\hat{\eta}_g^2 = 0.04$) but not a significant linear effect ($F(1, 159) = 2.9$, $p = .09$, $\hat{\eta}_g^2 = 0.02$). There were greater increases in ERS from the block preceding to the block following the feedback (*M* = 19.2, 95% CI [5.19, 33.2]) compared to the two pre-intervention or two post-intervention blocks (*M* = -3.22, 95% CI [-13.12, 6.69]). Therefore, the observed increases in lower alpha power from pre-feedback to post-feedback above can be attributed to the feedback, rather than just time-on-task. In the upper alpha frequency band, there was neither a significant linear ($F(1, 159) = 3.36$, $p = .07$, $\hat{\eta}_g^2 = 0.02$) nor quadratic effect ($F(1, 159) = 0.58$, $p = .45$, $\hat{\eta}_g^2 = 0$).

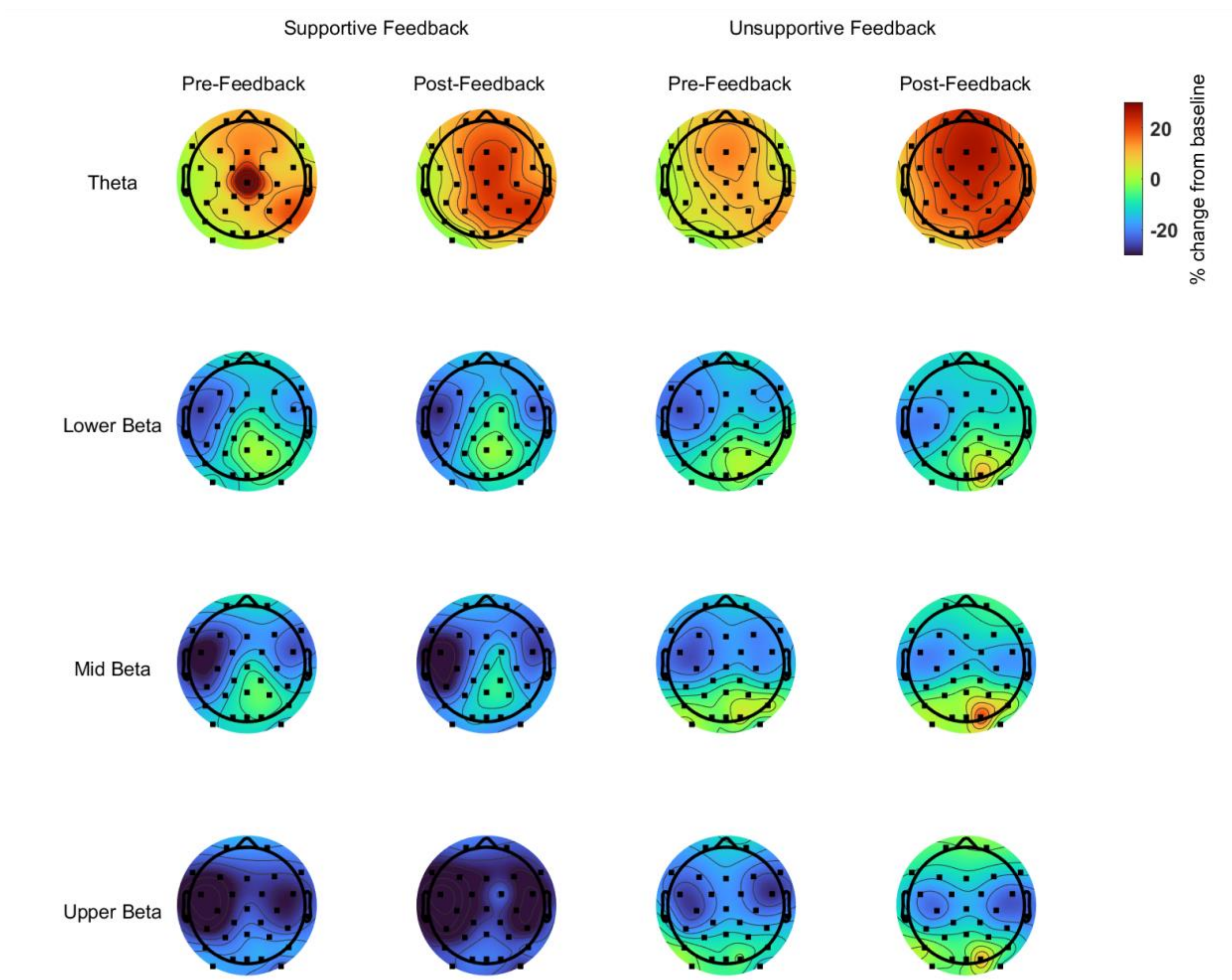
3.4 Exploratory Analysis of Beta and Theta Frequency Bands

We conducted exploratory analyses for task-related theta (4-7 Hz), as well as lower (12.5-16.5 Hz), mid (16.5-20 Hz), and upper (20-28 Hz) beta power changes. In the theta range, a significant main effect of PHASE ($F(1, 52) = 8.4$, $p = .005$, $\hat{\eta}_p^2 = 0.14$) showed that event-related synchronization (ERS) was greater post-feedback (*M* = 14.34, 95% CI [6.33, 22.35]) than pre-feedback (*M* = 6.48, 95% CI [0.88, 12.08]), mirroring results in the lower alpha band; see Figure 3.

In the lower beta range, no main effects or interactions reached significance. In the mid-beta range, we observed a significant PHASE by INTERVENTION interaction, ($F(1, 52) = 4.96$, $p = .03$, $\hat{\eta}_p^2 = 0.09$); see Figure 3. Pairwise comparisons revealed that for the unsupportive feedback condition only, event-related desynchronization (ERD) was greater pre-feedback (*M* = -12.35, 95% CI [-21.82, -2.87]) than post-feedback (*M* = -7.1, 95% CI [-16.67, 2.47], $p = .04$). Furthermore, post-feedback, ERD was greater for participants who received supportive feedback (*M* = -19.72, 95% CI [-28.94, -10.5]) than participants who received unsupportive feedback, although this effect was only marginally significant (p = .06). In the upper beta band, a main effect of INTERVENTION (F(1, 52) = 4.45, p = .04, $\hat{\eta}_p^2 = 0.08$) showed that ERD was greater for those in the supportive (M = -27, 95% CI [-36.56, -17.45]) than unsupportive (M = -12.53, 95% CI [-22.44, -2.62]) feedback condition; see Figure 3.

Figure 3

Task-related power changes across all electrodes pre- and post-feedback for the supportive and unsupportive feedback groups. Scalp maps reflect topographical distribution of the effects in the -1500 to -500 ms time window for theta (4-7Hz), lower beta (12-16.5Hz), mid beta (16.5-20Hz) and upper beta (20-28Hz) frequency bands.



4 Discussion

The present study investigated the behavioral and neural effects of critical feedback from a high-authority figure on engineering students' creative ideation. Feedback given by a professor midway through the experiment was predicted to either positively or negatively affect participants' creativity, depending on feedback type. Supportive feedback—in which the professor gave critical feedback but additionally expressed their high expectations and belief in the participant's ability to meet these expectations (G. L. Cohen et al., 1999; Yeager et al., 2014)—was predicted to positively impact creativity and elicit increased alpha ERS from pre- to post-feedback. Unsupportive critical feedback, in which participants were told that their ideas were "basic" and that they needed to try harder, was predicted to demotivate participants and thus negatively affect creativity outcomes and reduce alpha ERS from pre- to post-feedback. Exploratory analyses additionally investigated effects in the theta and beta frequency bands. The discussion will first review the behavioral and neural (alpha and theta frequency band) findings regarding changes from pre-feedback to post-feedback and will then interpret the beta frequency band findings regarding the feedback type (supportive versus unsupportive) manipulation.

In this study, we observed alpha synchronization during creative ideation over right-hemisphere posterior areas. This is in line with previous research, in which right-hemisphere posterior alpha ERS is robustly observed in tasks requiring creative ideation (Benedek et al., 2011; Fink & Benedek, 2014; Fink & Neubauer, 2006; Jauk et al., 2012; Jończyk et al., 2022, 2024; Rominger et al., 2019; Schwab et al., 2014). Alpha synchronization is associated with processes such as inhibitory processing and internally directed attention. In the context of idea generation, alpha ERS reflects an increase of these inhibitory processes needed to suppress uncreative ideas and internally directed attention required to generate creative ideas. Using a mid-experiment intervention in which participants received feedback on their performance, we further tested whether alpha synchronization during the creative ideation process is sensitive to social dynamics such as feedback from a high-authority figure. According to the feedback intervention theory, feedback makes an individual aware of the gap between their performance and a standard, which encourages individuals to implement better task strategies and can lead to improved performance (e.g., Kluger & DeNisi, 1996). We found that participants generated fewer but more original ideas after receiving feedback and, additionally, that alpha synchronization was greater post-feedback compared to pre-feedback, specifically in the lower alpha band. Follow-up analyses confirmed that lower alpha increases were greatest from the block immediately preceding to the block immediately following feedback, rather than a linear increase over all four blocks. Effects of feedback only in the lower (but not upper) alpha range suggests that feedback may affect general task-related processes, such as such as basic alertness (Benedek et al., 2011; Klimesch, 1999) more so than task specific demands (which are associated with upper alpha synchronization). Combined, these behavioral and neural results align with the feedback intervention theory and suggest that improved performance after critical feedback may be driven by increased lower alpha synchronization in the brain.

Our results also align with the emerging body of literature indicating that creative performance, and specifically neural oscillations during creative ideation, are affected by social dynamics (Fink et al,. 2011; Jończyk et al., 2022). Jończyk et al. (2022) for example, investigated creative ideation using a pre-post intervention paradigm similar to the current study and found increased alpha synchronization after inducing a stereotype threat in women engineers. These findings were interpreted to reflect that the invocation of a stereotype increased participant motivation and thus increased internally-directed attention and suppression of uncreative responses. This interpretation can be extended to the present study in which we investigate the influence of a different aspect of social dynamics, namely, feedback from a high-authority figure. Specifically, high-authority feedback may have boosted participants' motivation, leading to increased internally-directed attention, as reflected by both increased alpha synchronization post-feedback as well as corresponding post-feedback improvements in behavioral performance. Increased alpha ERS post-feedback also aligns with our exploratory findings in the adjacent theta band. Feedback has been shown to consistently elicit theta increases linked to performance monitoring and to top-down control mechanisms engaging attentional resources (Nurislamova et al., 2019). Increased theta power was observed post-feedback compared to pre-feedback; therefore participants may have also allocated greater attentional resources to generating more highly creative ideas after receiving feedback.

While we observed enhanced behavioral performance and increased alpha synchronization post-feedback compared to pre-feedback in line with our predictions, we did not find the predicted benefit for *supportive* or “wise” versus *unsupportive* feedback. Our manipulation check does show that participants rated the supportive professor more highly than the unsupportive professor on a scale of questions relating to the professor’s agreeableness, intelligence, reasonableness, supportiveness, motivation, and helpfulness. However, we did not observe a differential effect on behavioral outcomes or alpha synchronization for the supportive and unsupportive feedback conditions. While previous studies have highlighted the importance of “wise,” supportive feedback on students’ success, our results support the idea that feedback from a high-authority figure can motivate participants to try harder and produce more creative ideas, regardless of whether the feedback they received was supportive or unsupportive. Perhaps the two feedback types (supportive and unsupportive) were not sufficiently different from each other. However, given that the supportive and unsupportive scripts were based off of previous research (i.e. G. L. Cohen et al., 1999; Yeager et al., 2014) and that our post-task manipulation check indicated that participants did notice the nature of the intervention, this does not seem to be an adequate explanation. Instead, it is possible that any type of feedback (either supportive or unsupportive) may be beneficial for creative performance in engineering students, particularly for participants who are highly motivated by achievement. Fodor and Carver (2000) found that for high-achieving engineering students, performance on a task requiring creative responses to an engineering problem was greater for both those who received positive and those who received negative feedback compared to those who received no feedback at all. They explain that, compared to low-achieving individuals, high-achievers are highly responsive to information and feedback regardless of its valence and use feedback to improve task performance. While our participant recruitment did not particularly target high-achieving engineering students, there may have been a self-selection bias (i.e., students who are doing well enough in classes to have time to participate in a voluntary research experiment or students who were present in class during in-class recruitment). Indeed, questionnaire data indicate that our participants were highly committed to graduating with an engineering degree—on a scale from 1 (extremely uncommitted) to 10 (extremely committed), their mean response was 9.41.

In exploratory analyses of the lower, mid, and upper beta bands, we observe task-related *desynchronization*. Beta desynchronization has been associated with creative thinking (e.g., Jończyk et al., 2024; Mazza et al., 2023; Sheth et al., 2009). Mazza et al. (2023), for example, found greater beta desynchronization (along with typical alpha synchronization) during a divergent thinking AUT task compared to convergent thinking task. They interpret their findings to reflect greater internally directed attention during divergent thinking. In another study, Jończyk et al. (2024) found greater beta desynchronization when participants generated unusual uses of everyday objects in their second rather than native language, indicating lower cognitive effort associated with inhibition of ideas irrelevant to the task in the second rather than native language or, alternatively, difference in metacognition between languages, whereby participants would be less critical of their ideas in their second rather than native language. These findings align with the overall task-related beta desynchronization we observed in all three beta bands.

Interestingly, although there were no effects of feedback type (supportive versus unsupportive feedback) in the alpha band, exploratory analyses revealed significant effects of feedback type in the mid and upper beta bands. In the mid-beta range, there is an interaction between phase and intervention type, revealing that for those in the *unsupportive* condition, there is a significant *increase* in beta power (synchronization) from pre-feedback to post-feedback. Beta synchronization has been associated with cognitive processes related to active mental processing and cognitive effort (Engel & Fries, 2010; Spitzer & Haegens, 2017). Greater beta power has also been linked to a variety of emotional states including depressed mood (Begić et al., 2011; Grin-Yatsenko et al., 2009), rumination (Kühn et al., 2014), low positive affect (Abid et al., 2025), as well as stress and anxiety (Hein & Herrojo Ruiz, 2022; Palacios-García et al., 2021). Palacios-García et al. (2021) for instance, reported that participants in a stress condition showed a significant increase in beta power during an attentional task relative to a no-stress control group, and beta band activity was positively correlated with self-reported state anxiety and heart rate activity. Results were interpreted to indicate that individuals in the stressed condition turned their attention away from the task (to the threatening experience), then redirected attention to the task, increasing top-down control as evidenced by beta band activity. Here, we find that participants in the *unsupportive* feedback condition show greater mid-beta synchronization post-feedback, relative to participants with supportive feedback. Following the above interpretation, participants in the unsupportive condition may have been more focused on the feedback, therefore requiring more top-down control to redirect attention to the task, as evidenced by increased beta activity. In other words, negative feedback may have triggered compensatory cognitive efforts to analyze mistakes or poor performance and adjust strategies in the post-feedback phase of the task. Regardless of the explanation, differences in beta-activity between the supportive and unsupportive group did not translate to behavioral outcomes—recall there were no differences in idea originality or fluency between participants in the supportive or unsupportive feedback conditions.

Some limitations of this study need to be addressed. We conducted a manipulation check to validate that participants perceived the feedback provided by the professor in the supportive feedback condition to be different from that provided by the professor in the unsupportive feedback condition. This manipulation check indeed confirmed that the professor in the supportive feedback condition was rated significantly higher on a summed agreeable, intelligent, relatable, supportive, motivating, and helpful score than the professor in the unsupportive feedback condition. However, we did not measure how participants perceived and processed the *contents* of the feedback and whether, and if so how, they used the feedback they received to further improve their performance. Strijbos et al. (2021) provide an extensive review of questionnaires that have been developed and validated to measure how students perceive and process the contents of feedback provided by an external resource (e.g., teacher, peer, computer) and how students use the contents of the feedback to further improve their performance. Examples of these instruments are the Feedback Perceptions Questionnaire (Strijbos et al., 2010), Assessment Feedback Questionnaire (Lizzio & Wilson, 2008), Instructional Feedback Orientation Scale (King et al., 2009), and Feedback Orientation Scale (Linderbaum & Levy, 2010); see Strijbos et al. (2021) for a detailed overview of the subscales and psychometric quality indices of these and related questionnaires. Additionally, our sample size of 54 (28 in the supportive feedback condition, and 26 in the unsupportive feedback condition) may be considered on the low end, but is comparable to sample sizes in related studies (e.g., Händel et al., 2024; Jauk et al., 2012; Jończyk et al., 2022). Based on the confidence intervals and the rule of thumb for partial eta squared effect sizes (small effect = .001, medium effect = .004, large effect = .14), we observed medium to large effects for both behavioral and EEG measures, indicating that our study is sufficiently powered.

Additionally, our experimental design leaves some unanswered questions. One relates to our design implementation of supportive versus unsupportive feedback. To maximize the likelihood of detecting any differences between supportive ("wise") feedback and unsupportive feedback, the supportive feedback was delivered by a female professor and the unsupportive feedback by a male professor. Given ample evidence that females are, stereotypically, perceived to be more supportive and use more affiliative speech (for review, see Leaper & Ayres, 2007) than men, we assumed that a female professor delivering the supportive feedback would boost its impact. Even though our manipulation check confirmed that the female professor was indeed rated more highly (on a summed measure of agreeableness, intelligence, relatability, supportiveness, motivation, and helpfulness) than the male professor, the specific nature of the feedback (supportive or unsupportive) did not further quality the overall impact of feedback on the behavioral measures of creative thinking or alpha oscillations, as predicted. However, based on our design, we cannot rule out that professor gender may have somehow neutralized effects of supportive versus unsupportive feedback. Future research that seeks to specifically test the relation between gender of the feedback provider and nature of feedback may want to implement the full 2 (gender: male, female) by 2 (feedback: supportive, unsupportive) between-subjects design. A second point is that our study did not include a separate sample of participants to serve as a control condition in which no feedback was delivered. However, our study did include two blocks pre-feedback, which effectively serves as a within-groups control; linear effects which would indicate significant changes from Block 1 to Block 2 were not found for either behavioral or neural measures. Even so, future research may consider including a between-participants control condition in which no feedback is given.

To conclude, this study is the first to our knowledge to investigate the role of feedback on creative thinking and brain dynamics. Our results provide novel evidence that critical feedback from a high-authority figure—regardless of whether that feedback is supportive or unsupportive—is associated with increased originality of creative responses as well as increased alpha synchronization during idea generation in undergraduate engineering students. Critically, unsupportive relative to supportive feedback induced increased beta oscillation power, possibly reflecting increased cognitive stress and internally directed attention required to adjust performance after unsupportive feedback. Feedback is just one of many potential social variables that may play a role in affecting creative ideation. Future research could explore other avenues including the extension of these findings to feedback from peers rather than high-authority figures, or the effect of feedback in a group setting that may more closely reflect ideation as it occurs in education or workplace environments.

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Data Availability Statement: The data that support the findings of this study, R code used for data analysis, and reports for supplemental analyses are openly available in OSF at https://osf.io/gmr58/?view_only=2114347413ce47499deb8562292078a2.

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Appendix A

Scripts for feedback given during mid-experiment feedback intervention.

Supportive Feedback Script

Hi, my name is XX and I am the head of this lab. Thank you very much for participating in our research. I really appreciate the time you are spending with us, and that you are helping us with our study. It is obvious to me that you are taking this task seriously. I always check in half-way through the experiment to see how things are going. This task is pretty challenging and finding alternative uses for these objects and ideas for the hypothetical situations is not something we do every day. But I would like to encourage you to think more outside the box and think of more creative ways that these objects can be used in new ways, and in new situations. So try to think more creatively and try to come up with truly novel ways objects can be used, or novel ways the hypothetical situations could unfold. When I was checking the EEG recordings in the experimenter room outside, I heard that you came up with some very creative uses of a chair during the practice block and for the ice age situation. This tells me that you are a creative thinker and have the mind-set to come up with some very original and new ways these objects can be used. I also think you are capable to think of novel ways hypothetical situations can unfold. Based on what I heard that you said earlier, I have high expectations of you and I know you can reach them. So let's move on to the next series of objects and situations!

Unsupportive Feedback Script

Hi my name is Dr. XX and I am affiliated with this lab. Thank you very much for participating in our research. I appreciate the time you are spending with us, and that you are helping us with our study. And thank you for your efforts. I always check in half-way through the experiment to see how things are going. This task is difficult, and finding alternative uses for these objects and ideas for the hypothetical situations is not something we do every day. Just please try to think a bit harder, and let your brain work a bit more. I just wanted to give you some feedback and check in with you. When I was checking the EEG recordings in the experimenter room outside, I heard that you came up with some pretty basic ideas for a chair during the practice block and for the single continent situation. So, I was basically coming in to share my thoughts and to let you know that this is a hard task. Actually, many of the engineering and psychology students we have tested so far find it hard, so I guess you are in good company. But please try to do a bit better in the second half, and let your mind work harder. And please try to speak clearly because this is important for the recordings. So let’s move on to the next series of objects and situations!

Appendix B

Scripted conversations describing interactions with an academic mentor during EEG cap preparation.

Supportive Feedback Condition

Speaker 1: How is your dissertation project going? The result section sounded like you got a lot of edits and changes to be made…

Speaker 2: Yes, well, XX, my advisor, wants me to produce the best work possible, so that we go above the basic standards in our publications. It's tough because our approach is truly novel and not a lot of labs are using these analyses in child data, and she's setting the standards for it high. It's important that people believe what we're talking about.

Speaker 1: Right, if you don't persuade people in the results section then they won't buy the rest of the paper. No wonder she wants it done thoroughly.

Speaker 2: Yeah… I was feeling not so great about all those new analyses, but XX pointed out that I had already succeeded in doing these analyses with adult data, so this should be possible too. It was good to know that Janet thought I could do it!

Speaker 1: Yeah, having high standards seems like a sign of good faith especially if she's talking about the other things you've done before. And even if there are a lot of new things to work on at least you know you have the capacity to pull it off.

Unsupportive Feedback Condition

Speaker 1: How is your dissertation project going? The result section sounded like you got a lot of edits and changes to be made…

Speaker 2: Yes, well, XX, who is the outside reader, wants me to get my work done so that we can get it published as quickly as possible. It's tough because our approach is truly novel and not a lot of labs are using these analyses in child data, and he just wants to get it out. But, it's important that we can show that I have publications and I guess so people can read our work. It's important that people know what we've worked on.

Speaker 1: Right, and if you don't publish that doesn't look great for the lab or for him. It's important for him to get papers out the door for the publication record and for science.

Speaker 2: Yeah… still, I was feeling not so great about all those new analyses. He made a lot of comments were basically like "I don't get it" / "it's not clear" but I don't really know how to fix it from there.

Speaker 1: That's rough. Getting work out is good, though. It just takes a while when the feedback is not so specific about what he wants you to change. But you'll be done soon enough and, like, you'll figure out what he wants you to write eventually.

Appendix C

Post-Task Manipulation Check

Response scale: 1 (strongly disagree) 2 3 4 5 6 7 (strongly agree)

1. The professor who came in was agreeable.
2. The professor who came in was intelligent.
3. The professor who came in was relatable.
4. The professor who came in supportive.
5. The professor’s comments motivated me to perform at my best.
6. The professor’s comments helped me to come up with novel solutions.